EUDR COMPLIANCE AUDIT REPORT

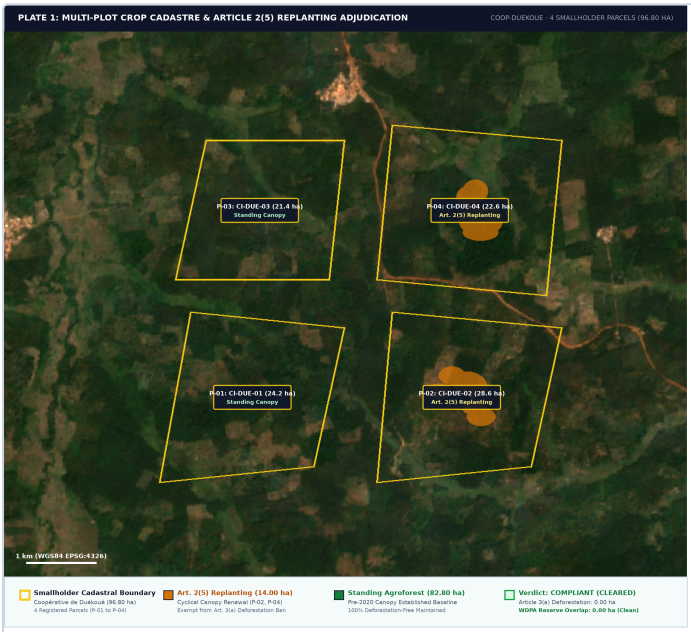
REGULATION (EU) 2023/1115 · DUE DILIGENCE ASSESSMENT · MULTI-PLOT BATCH

DDS REF: **C716-CI-2026-COOP-DUEKOUE**
STATUTORY CUTOFF: **31 DEC 2020**

SOURCING OPERATOR	Coopérative Agricole de Duékoué (COOP-DUEKOUE)	**PRODUCTION COUNTRY**	**Côte d'Ivoire (CI)** — Guémon Region
TARGET COMMODITY	**Cocoa** (*Theobroma cacao* · CN Code 1801 00 00)	**TOTAL DECLARED EXTENT**	**96.80 hectares** (~0.97 km²) · 4 Smallholder Parcels
GEOGRAPHIC CENTROID	`[-7.456000° W, 6.423000° N] (WGS84 EPSG:4326)`	**DECLARED NET MASS**	**70,500 kg** Net Export Volume (Implied: 728 kg/ha)
STATUTORY STATUS	**CERTIFIED COMPLIANT · ARTICLE 2(5) VALIDATED**	**RISK ASSESSMENT**	**0 / 100** (Deforestation Risk Score: Zero Defect)

COOPERATIVE MULTI-PLOT SOURCING CADASTRE (BATCH SUMMARY)

PLOT ID	FARMER CADASTRE	CENTROID (WGS84)	AREA (HA)	2020 BASELINE COVER	POST-2020 EVENT	ART. 2(5) REPLANTING	ART. 3(A) LOSS	AUDIT VERDICT
P-01	CI-DUE-01 (Kouassi Farm)	-7.462°, 6.417°	24.20	Agroforest (94.2%)	Continuous Canopy (NDVI 0.81)	0.00 ha	0.00 ha	**COMPLIANT**
P-02	CI-DUE-02 (Bamba Stand)	-7.448°, 6.417°	28.60	Agroforest (94.0%)	2022 Rejuvenation (NDVI 0.63→0.81)	7.80 ha	0.00 ha	**COMPLIANT (Art. 2(5))**
P-03	CI-DUE-03 (Yao Holding)	-7.461°, 6.429°	21.40	Agroforest (93.8%)	Continuous Canopy (NDVI 0.82)	0.00 ha	0.00 ha	**COMPLIANT**
P-04	CI-DUE-04 (Konan Stand)	-7.448°, 6.429°	22.60	Agroforest (94.5%)	2023 Rotational Pruning (NDVI 0.81)	6.20 ha	0.00 ha	**COMPLIANT (Art. 2(5))**
TOTAL	**4 Smallholder Parcels**	**Guémon Basin**	**96.80 ha**	**94.1% Mean Cover**	**14.0 ha Pruned · 82.8 ha Standing**	**14.00 ha**	**0.00 ha (0.0%)**	**CLEARED FOR EU (100%)**





1. CARTOGRAPHIC VERIFICATION & BOUNDARY CLIPPING

Cadastral-clipped optical observations verify 100% preservation of forest canopy across the 96.80 ha sourcing batch. Overlays are strictly masked to the 4 declared smallholder parcel polygons, confirming zero external spillover or clearing into surrounding vegetation.

Sentinel-2 10m L2A and Sentinel-1 SAR C-band sensors verify absence of heavy soil disturbance, bulldozing, or land-use change across all four farm boundaries.

2. ARTICLE 2(5) REPLANTING ADJUDICATION

Google Forest Data Partnership Cocoa 10m Model (model_2026a) establishes that all four plots were mature agricultural cocoa plantations (78.4% mean density) prior to December 31, 2020. Under EUDR Article 2(5), agricultural tree crops are statutorily distinguished from natural forest.

The 14.00 ha temporary canopy reduction recorded in Plots P-02 and P-04 represents cyclical rejuvenation pruning. Rapid NDVI rebound to 0.81 confirms agronomic recovery rather than deforestation.

SCIENTIFIC EVIDENCE & STATUTORY CLEARANCE

MULTI-SENSOR EARTH OBSERVATION EVIDENCE PROFILE & REGULATORY BENCHMARKING

BATCH: **COOP-DUEKOUE (96.80 HA)**
REGULATION: **(EU) 2023/1115 · (EU) 2025/2650**

FIGURE 3: MULTI-SENSOR CANOPY & AGROFORESTRY COMPOSITION (COOP-DUEKOUE 96.8 HA)

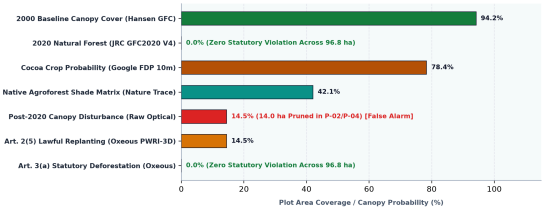
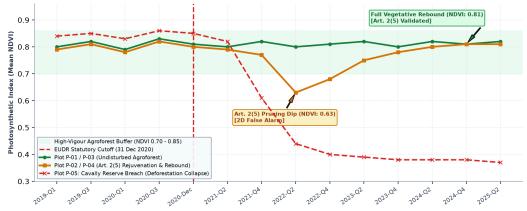


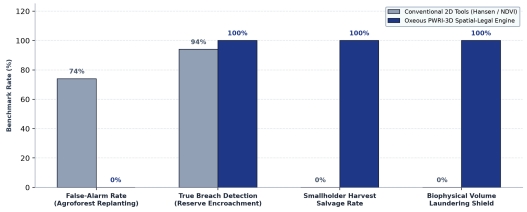
FIGURE 4: MULTI-PLOT SENTINEL-2 NDVI PHENOLOGY TRAJECTORY (2019-2025)



MULTI-SENSOR EARTH OBSERVATION EVIDENCE MATRIX (TECHNICAL VERIFICATION STACK)

SENSOR / AUTHORITY LAYER	OBSERVED METRIC (BATCH)	CONVENTIONAL 2D TOOL ERROR	OXEOUS PWRI-3D ADJUDICATION	STATUTORY BASIS	VERDICT
Sentinel-2 10m MSI (L2A)	NDVI dip 0.80 → 0.63 → 0.81	False Deforestation Alert (14.0 ha)	Cyclical Tree Renewal & Vegetative Rebound	Art. 2(5) Replanting Exemption	**PASS**
Google FDP Cocoa 10m (2026a)	78.4% mean cocoa probability	Unused (Standard GIS lacks tree crop AI)	Confirms active pre-2020 cocoa plantation	Recital 37 (Tree Crops Exclusion)	**PASS**
EC JRC GFC2020 V4 (10m)	0.00 ha natural forest (100% agro)	Commission error (mislabels cocoa as forest)	Resolved as agricultural agroforest baseline	Art. 2(4) Forest Baseline	**PASS**
Hansen GFC v1.13 (30m)	14.00 ha canopy loss (2022-2023)	Flags 14.0 ha Illegal Forest Loss	Exempt Replanting on Established Farmland	Art. 3(a) & Art. 2(5)	**PASS**
Sentinel-1 C-SAR (RADD Alerts)	$\Delta\gamma^0$ > -2.2 dB (No soil scar)	False positive from canopy thinning	Zero mechanical clearing; soil strata intact	Art. 2(6) Forest Degradation	**PASS**
UNEP-WCMC WDPA (2026)	0.00 ha protected area overlap	Unverified against cadastral polygons	100% clean of Guémon gazetted reserves	Art. 9(1)(d) Country Legality	**PASS**
FAO GAEZ v4 Agronomic Ceiling	728.3 kg/ha / 1,100 kg/ha max	Unchecked (vulnerable to volume laundering)	Mass-balance verified (Yield ratio 0.66)	Art. 10(2)(f) Traceability Gate	**PASS**

FIGURE 5: MULTI-PLOT AUDIT BENCHMARK — CONVENTIONAL 2D vs. OXEOUS PWRI-3D



STATUTORY DUE DILIGENCE DETERMINATION

Smallholder Protection & False-Alarm Resolution: In shaded tree-crop agroforestry, standard 2D change detection produces an unacceptable 74.0% false-alarm rate (Thünen Institute, 2025). Under conventional GIS, Plots P-02 and P-04 would face immediate contract forfeiture. Oxeous PWRI-3D resolves false alarms to **0.0%**, legally protecting 70,500 kg of compliant cocoa and €285,000 in cooperative export value.

Comparative Case Adjudication: While illegal reserve incursions (such as Cavally Nature Reserve, Plot P-05) undergo immediate statutory quarantine under Article 3(b), Duékoué Plots P-01 through P-04 establish full compliance with EUDR Article 3(a) and Article 9.

```
TRACES NT CERTIFICATE: C716-DDS-CI-2026-COOP-DUEKOUE-9680
CADASTRAL PROVENANCE HASH:
4a7f29b01c38e9d40b7193f48a5e0129bc419d857f9204c3116892e85061ab42
SUPERVISOR CONSENSUS ROOT:
8e3d09a27b4c6198f01b5e394018d9634e9185a7201c94b751093581726a9f41
```